

Network Devices -- Guided Notes ANSWER KEY

Unit tp-2.7 | CompTIA Network+ N10-009 Obj. FC0-U71 2.7 | N10-008 FC0-U71 2.7

For instructor use / distribute after students complete the notes.

Question 1

A _____ operates at Layer 2 (Data Link) and forwards traffic based on _____ addresses. It creates a _____ collision domain per port, isolating each connected device. This is the primary device used to connect computers within a local area network (LAN).

Answer: Switch; MAC; separate.

Question 2

A _____ operates at Layer 3 (Network) and forwards traffic between different networks based on _____ addresses. The router in a home connects the local network to the _____ (Internet). It is the default _____ for devices on the LAN.

Answer: Router; IP; Internet (ISP); gateway.

Question 3

A WAP (Wireless Access Point) provides _____ network connectivity by broadcasting an _____ using a radio signal. A home wireless router combines a router, switch, and _____ into one device. An enterprise may have _____ WAPs managed by a central wireless controller.

Answer: Wi-Fi (wireless); SSID; WAP (wireless access point); many.

Question 4

A _____ (cable/DSL modem) converts the signal from an ISP (Internet Service Provider) into a format the local network can use. _____ modems use a coaxial cable connection. _____ modems use a telephone line. The word modem comes from _____ and _____ because it converts between analog and digital signals.

Answer: Modem; cable; DSL; modulate; demodulate.

Question 5

A _____ is a network security device that monitors and controls incoming and outgoing traffic based on _____ rules. A hardware firewall is a dedicated _____ device protecting the entire network. A software firewall runs on an _____ device and protects only that machine.

Answer: Firewall; policy/security; network; individual (host).

Question 6

A _____ (network hub) is a legacy Layer 1 device that broadcasts every packet to _____ ports. This creates a _____ collision domain and wastes bandwidth. Hubs have been replaced by _____ in all modern networks because switches forward traffic only to the intended recipient.

Answer: Hub; all; shared (single); switches.

Question 7

PoE (Power over Ethernet) allows a _____ cable to carry both data and _____ power. This is used to power devices such as _____, _____, and _____ (list at least two) without requiring a separate power outlet at each device location.

Answer: Ethernet (Cat5e/Cat6); electrical (DC); IP cameras, VoIP phones, wireless access points (any two).

Question 8 -- Real World Application

REAL WORLD: A school IT team is upgrading a computer lab with 30 student computers. They currently have one old hub connecting all machines. They notice that when multiple students download files simultaneously, everyone's connection slows to a crawl. Explain why this is happening and how replacing the hub with a managed switch would solve the problem.

Answer: A hub operates at Layer 1 and broadcasts every frame out of every port. All 30 computers share one collision domain. When multiple computers transmit simultaneously, their signals collide, forcing CSMA/CD back-off retransmission, which dramatically reduces throughput. Each computer effectively divides the total bandwidth by 30. A managed switch operates at Layer 2 and maintains a MAC address table. It forwards each frame only to the correct destination port, giving each computer a dedicated collision domain. With a 1 Gbps switch, each of the 30 computers gets its own 1 Gbps link rather than sharing one. This eliminates collisions and allows simultaneous downloads at full speed.